

Error-Aware ML Potentials for Cost-Efficient Atomistic Simulations

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1 Motivation

Common atomistic ML models:

$$y_i = f(x_i) + \epsilon_i, \text{ where } \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- ✗ All conducted simulations have to be equally high quality.
- ✗ This is often infeasible for large datasets.

Error-informed Gaussian process Regression (GPR):

$$y_i = f(x_i) + \epsilon_i, \text{ where } \epsilon_i \sim \mathcal{N}(0, \sigma_i^2)$$

- ✓ Reduce data generation cost while keeping high predictive accuracy.
- ✓ Discretization errors can be estimated¹.

2 Objective

Develop an **error-informed** GPR model for predicting DFT quantities enabling accelerated and cost-efficient computational materials discovery.

3 Methods

Gaussian process²

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

Joint prior distribution of train $\mathbf{y}$ and test $\mathbf{f}_* = f(x_*)$:

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N} \left(\mathbf{0}, \begin{bmatrix} K(X, X) + \Sigma & K(X, X_*) \\ K(X_*, X) & K(X_*, X_*) \end{bmatrix} \right)$$

Noise covariance: $\Sigma = \sigma_n^2 I$ vs $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_N^2)$

Posterior distribution:

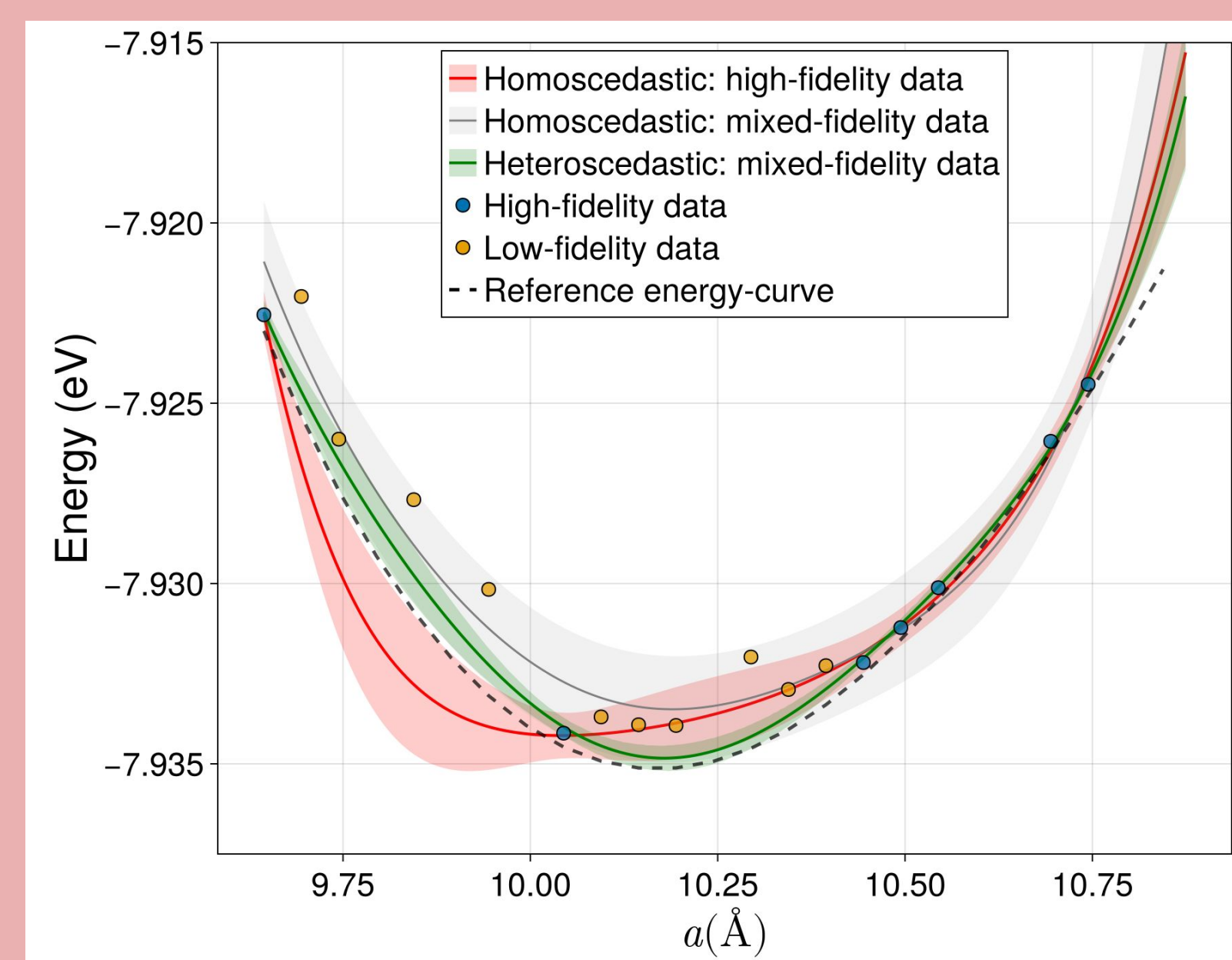
$$\mathbf{f}_* | X, \mathbf{y}, x_* \sim \mathcal{N}(\bar{\mathbf{f}}_*, \text{cov}(\mathbf{f}_*))$$

where

$$\bar{\mathbf{f}}_* = \mathbf{k}_*^\top (K + \Sigma)^{-1} \mathbf{y},$$

$$\text{cov}(\mathbf{f}_*) = k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^\top (K + \Sigma)^{-1} \mathbf{k}_*.$$

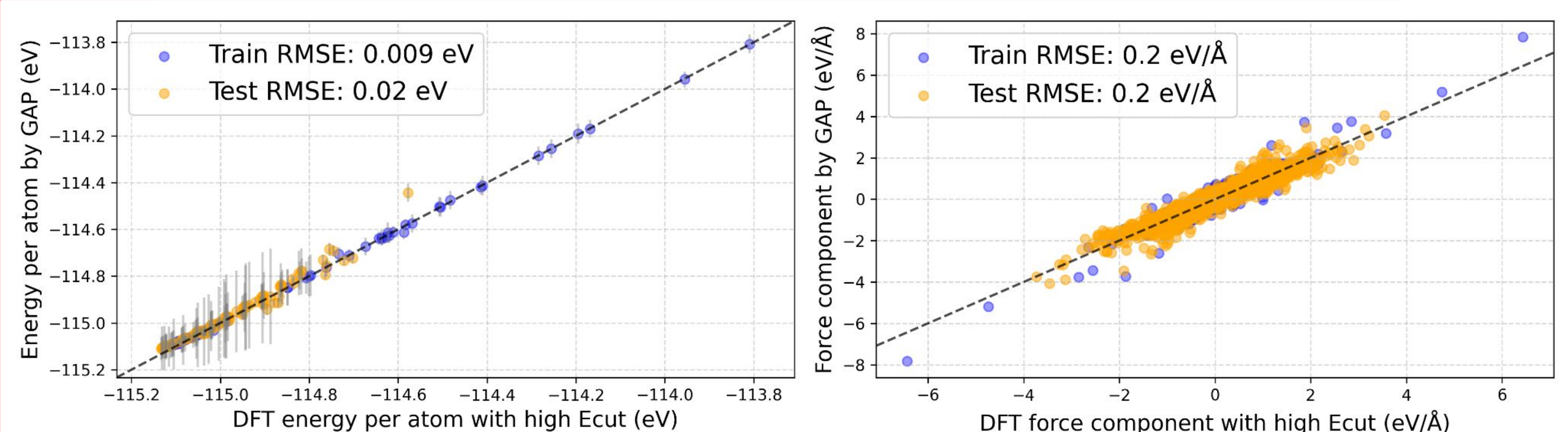
4 Results: Energy-Volume (EV) curve



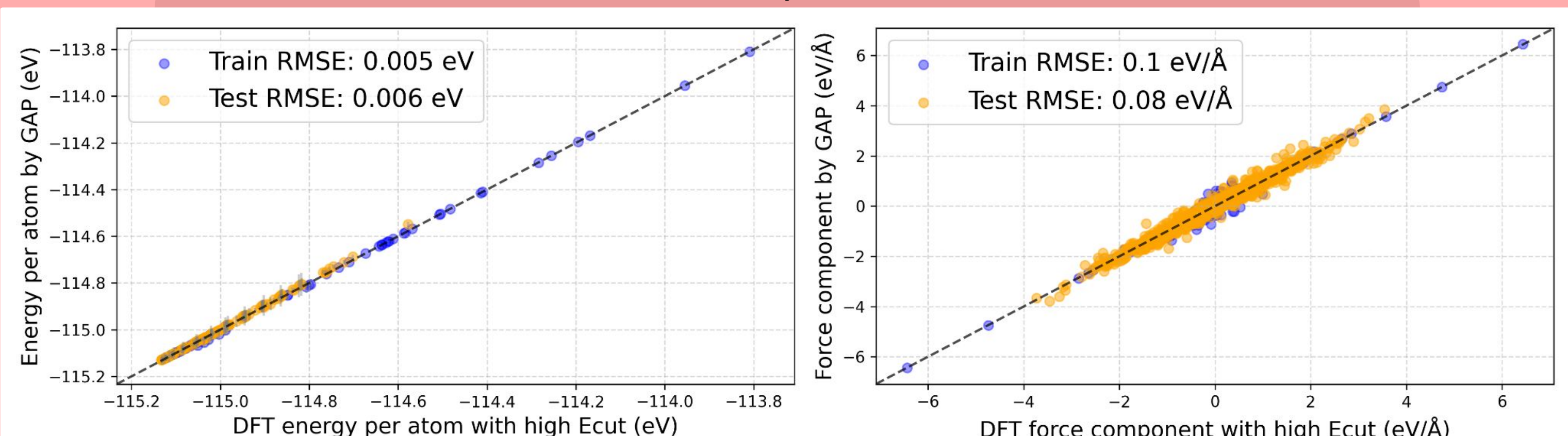
Approximation with a GPR for a mixed dataset with varying plane-wave cutoff.

5 Results: Error-aware GAP framework

Standard GPR: multi-fidelity dataset, Ecut = 10, 18 and 26 Ha



Error-informed GPR: multi-fidelity dataset, Ecut = 10, 18 and 26 Ha

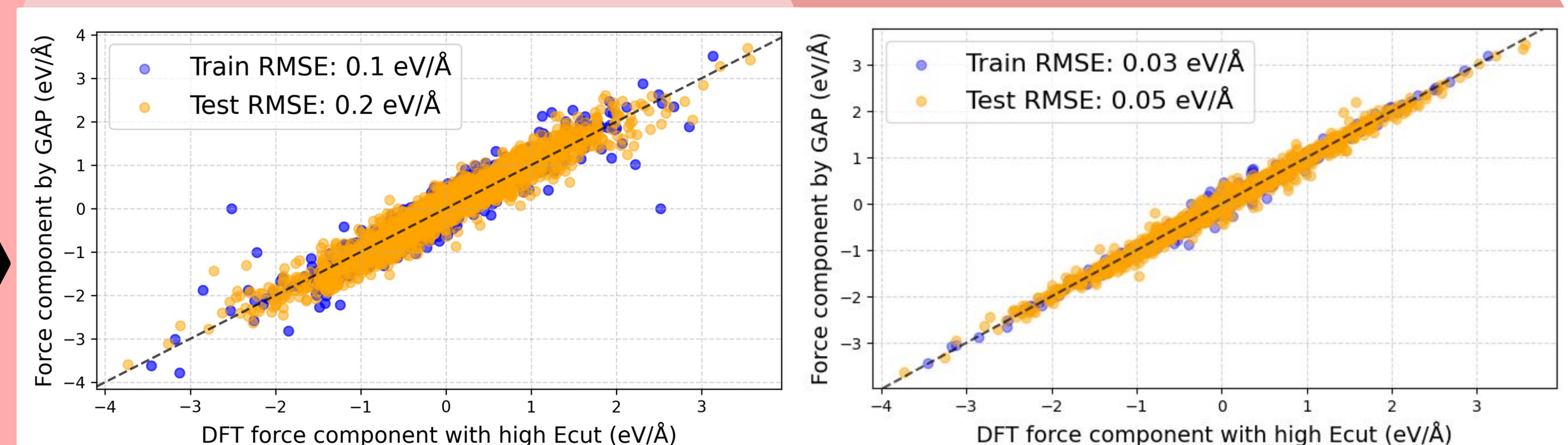


- ✓ Lower RMSE on energies: 0.02 → **0.006 eV**
- ✓ Lower RMSE on forces: 0.2 → **0.08 eV / Å**

Low-fidelity dataset, Ecut = 10 Ha: 2103 environments

Standard GAP

Error-informed GAP



- ✓ Lower RMSE on forces: 0.2 → **0.05 eV / Å**

6 Conclusion

We integrated **error-aware modeling** into the GAP framework, enabling accurate **uncertainty quantification** for both energy and force predictions, and demonstrated improvements in **prediction accuracy** on multi- and low-fidelity datasets of Silicon structures, while reducing total **data generation cost**.

References

1. Eric Cancès, Geneviève Dusson, Gaspard Kémlin, and Antoine Levitt, 2022. Practical error bounds for properties in plane-wave electronic structure calculations. SIAM Journal on Scientific Computing.
2. Rasmussen, C.E., Williams, C.K.I., 2006. Gaussian Processes for Machine Learning. The MIT Press.